

Mohamed Amine Bayar,

PhD | Independent Biostatistics

Consultant | Oncology & Hematology

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English, French & Arabic

Profile

- PhD-level biostatistics consultant with 13+ years across the full clinical development lifecycle — from Bayesian Phase I dose-escalation through Phase III pivotal design, high-stakes analysis milestones, and regulatory submission.
- A profile combining frontline industry experience (Johnson & Johnson, Novartis), academic research depth (PhD Paris-Saclay, 15+ publications, 1,200+ citations), hands-on ML/AI expertise applied to clinical and genomic data (co-invented patent WO2018002385A1), and advanced programming capabilities (R/Shiny, Python, SAS). Direct FDA and EMA interactions on pivotal trial design and regulatory strategy.
- Core expertise in oncology and hematology (multiple myeloma, CML, solid tumors — 8+ disease areas); methodology and regulatory experience transferable across therapeutic areas.

Technical Expertise

- **Statistical Methodology:** Survival analysis (Cox, competing risks, frailty models); linear, logistic, and mixed-effects models; Bayesian dose-escalation (CRM, BLRM, BOIN); group sequential and adaptive designs (seamless Phase II/III, platform trials); multiple testing procedures; Estimand framework (ICH E9(R1)); sample size calculation and event re-estimation; penalized regression (Lasso, Ridge, Elastic Net); PKPD and exposure-response modeling.
- **Programming — R:** tidyverse, ggplot2, RShiny, RMarkdown; OncoBayes2 (Bayesian BLRM dose-escalation); rpact, gsDesign (group sequential designs); survival, cmprsk (time-to-event); automated TFL pipeline development.
- **Programming — Python:** pandas, NumPy, scikit-learn (ML/AI); lifelines, scikit-survival (survival analysis); XGBoost, SHAP (predictive modeling and interpretability).
- **Programming — Other:** SAS (Base, Stat, Macro), SQL; nlmixr (PKPD/tumor growth modeling, R).
- **ML/AI:** Predictive modeling and prognostic signature development; variable selection in high-dimensional genomic spaces; internal and external model validation (cross-validation, bootstrap); co-invented patent applying ML to oncology outcomes (WO2018002385A1).
- **Medical Writing:** SAPs, TFL shells, CSRs, SCS/SCE, briefing documents, health authority responses, IBs.

Work Experience

BioSpark

Independent Biostatistics Consultant

Provides independent statistical consulting across the full drug development lifecycle, from early-phase design through regulatory submission. Developing expertise at the intersection of classical biostatistics and AI/ML methodologies.

Johnson & Johnson

Statistician — Oncology & Hematology, 3 years & 2 months

Remote, France

Apr 2026 – Present

Remote, France

Feb 2023 – Mar 2026

Trial statistician across four oncology programs (multiple myeloma and NSCLC), spanning statistical design, 10+ high-stakes analysis milestones, and regulatory submission.

- **Analysis delivery (teclistamab, ciltacabtagene autoleucel):** Led statistical delivery of multiple analysis packages — interim analyses, safety snapshots, and primary analyses — across two cutting-edge myeloma programs in Phase II multi-arm settings: teclistamab (BCMA×CD3 bispecific); and ciltacabtagene autoleucel (BCMA-directed CAR-T). Provided CRO statistical oversight on two studies and contributed to onboarding and mentoring of new statisticians joining the team.
- **Statistical design and study conduct (ramantamig):** Served as lead statistician for Phase Ib and Phase III design on a first-in-class BCMA×GPRC5D×CD3 trispecific — including BOIN dose-escalation design and simulation, Phase III event re-estimation, interim efficacy analyses, OS harm monitoring, and direct FDA/EMA interactions from concept through regulatory alignment (EOP2 briefing book submission and governance activities).
- **Regulatory submission (amivantamab):** Contributed to the submission package for amivantamab in NSCLC — extending regulatory experience beyond myeloma.
- Authored Protocols statistical sections, SAPs, TFL shells, CSRs, and regulatory briefing documents; prepared written responses to FDA/EMA health authority queries.

Novartis

Paris, France

Statistician — CML Portfolio (Scemblix), 3 years 3 months

Nov 2019 – Jan 2023

- **Phase II statistical delivery (asciminib):** Led full statistical delivery for a 4-arm Phase II study — from SAP development through database lock, primary analysis, and reporting — forming part of the integrated evidence base supporting the approved label.
- **Pediatric regulatory program (PIP/iPSP):** Led statistical methodology for a Phase Ib adult-to-pediatric bridging study developed under the joint EMA/FDA pediatric framework (Paediatric Investigation Plan and initial Pediatric Study Plan) — a mandatory regulatory requirement for novel oncology agents. Implemented Bayesian BLRM dose-escalation using the OncoBayes2 R package, including prior calibration from adult data and simulation to support dose recommendation in a pediatric population.
- **Pharmacometrics rotation (formal secondment):** Completed a 7-month full-time rotation within the Novartis pharmacometrics team — nlmixr R package.

Gustave Roussy Cancer Center

Paris, France

Statistician — Solid Tumors, 6 years 9 months

Mar 2013 – Oct 2019

Trial statistician on 7 clinical trials (Phase I to III) and multiple translational research projects, with responsibility for trial delivery, methodological research, and scientific publication.

- **Trial delivery and analysis milestones:** Led statistical activities on named programs including POP, CHIPASTIN, HPVRX, MELIPI RX, PACS04, IGRT-P, and others — spanning breast (TNBC, HR+, metastatic, early), ovarian, melanoma, prostate, and pediatric oncology. Delivered full analysis lifecycles including interim, primary, and final analyses.
- **Translational research & prognostic modeling:** Developed and validated multiple prognostic models across tumor types — including three sarcopenia prognostic models, a risk stratification model in pediatric intracranial ependymoma, and a survival prognostic model in second-line metastatic renal cell carcinoma.
- **Precision medicine & platform trials (SAFIR02, MAPPYACTS):** Substantial contributor to two landmark precision medicine genomic screening platforms — delivering fully automated TFL pipelines for IDMC meetings. PhD research (Part 3) focused on optimal statistical analysis strategies for platform trials such as SAFIR02.
- **Co-invented patented ML pipeline:** Built and validated a genomic prognostic signature for TNBC (Patent WO2018002385A1) — owning the full pipeline from raw gene expression extraction through data normalization (fRMA), cross-platform merging, predictive modeling, and internal/external model validation; published in *Annals of Oncology*.
- **Methodological research & academic output:** Generated 15+ peer-reviewed publications (1,200+ citations) across *Annals of Oncology*, *JCO*, *CCR*, *Statistics in Medicine*; Invited visiting scientist at Mayo Clinic, hosted by Prof. Daniel J. Sargent. Taught biostatistics and public health to undergraduates at Paris Sud University.

Education

Paris Saclay University

Paris, France

PhD in Public Health — Major Biostatistics

Oct 2016 – Nov 2019

- Topic: Long-term evaluation of randomized Phase III clinical trials in oncology with rare diseases and biomarker-based subtypes.
- Developed novel methodology for randomized trials with seamless Phase II/III, platform trials, and interim analysis strategies; and produced 3 first-author publications in *Statistics in Medicine*, *Statistical Methods in Medical Research*, and *Computer Methods and Programs in Biomedicine*.

ESSAI — Université de Carthage

Tunis, Tunisia

Engineer in Statistics

Sep 2010 – Jun 2013

Diplôme National d'Ingénieur — Statistique et Analyse de l'Information

Intellectual Property & Innovation

Patent: WO2018002385A1 — hands-on ML application to medicine: built and validated a genomic signature to predict prognosis in Triple Negative Breast Cancer, owning the full pipeline from raw gene expression data to a clinically validated predictive model. <https://patents.google.com/patent/WO2018002385A1/en>

Selected Publications

2020: Mohamed Amine Bayar, Gwénaél Le Teuff, Franz Koenig, Marie-Cécile Le Deley, Stefan Michiels. *Group sequential adaptive designs in series of time-to-event randomized trials in rare disease: a simulation study*. Statistical Methods in Medical Research, Jun. 2020. <https://pubmed.ncbi.nlm.nih.gov/31354106>

2016: Mohamed Amine Bayar, Gwénaél Le Teuff, Stefan Michiels, Daniel J Sargent, Marie-Cécile Le Deley. *New insights into the evaluation of randomized controlled trials for rare diseases over a long-term research horizon: a simulation study*. Statistics in Medicine, Aug. 2016. <https://www.ncbi.nlm.nih.gov/pubmed/27027783>

2018: C Criscitiello, **MA Bayar**, G Curigliano, FW Symmans, C Desmedt, H Bonnefoi, B Sinn, G Pruneri, Vicier, Pierga, Denkert, S Loibl, C Sotiriou, S Michiels, F André. *A gene signature to predict high tumor-infiltrating lymphocytes after neoadjuvant chemotherapy and outcome in patients with triple-negative breast cancer*. Annals of Oncology, Jan. 2018. <https://www.ncbi.nlm.nih.gov/pubmed/29077781>

Full list: 15+ papers, 1,200+ citations: <https://scholar.google.com/citations?user=u6Ns41gAAAAJ&hl=en>